Data Storytelling

Explaining data through narrative and interaction

Information Visualization · Lecture 13 · Week 7

Five parts of a data story

01	**Foundations**	Exploration and explanation
02	**Narrative sequence**	Three acts, four moves, and pacing
03	**Attention**	Emphasis, annotation, and reading paths within a view
04	**Balance**	Author guidance, reader exploration, structures and genres
05	**Storyboarding**	Planning scenes, transitions, and reader choices

Exploration and explanation serve different tasks

Data storytelling combines data visualizations with a narrative: an organized sequence of explanations that helps readers understand a topic or finding.

Exploration

The analyst varies views, asks new questions, and searches for patterns.

Typical experience: an open-ended analysis tool.

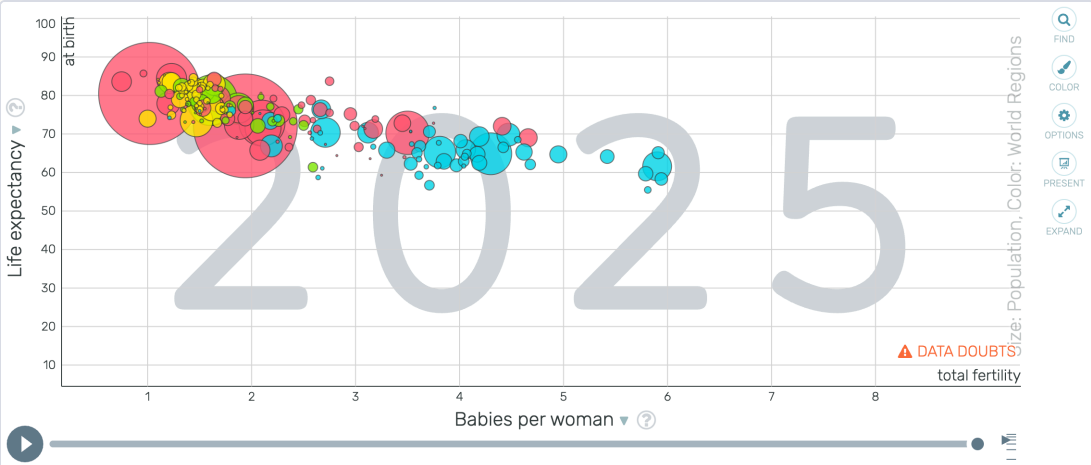
Explanation

The author arranges views and commentary to develop an understandable account.

Typical experience: a guided data story.

Interactive data stories place reader choices within an explanatory sequence. See Segel & Heer (2010), introduction.

A chart can be explored or narrated



Gapminder Tools · Open original ↗



Hans Rosling / BBC News · Watch the film ↗

Tools: the reader chooses variables and comparisons.

Film: narration and animation introduce the chart and move through time.

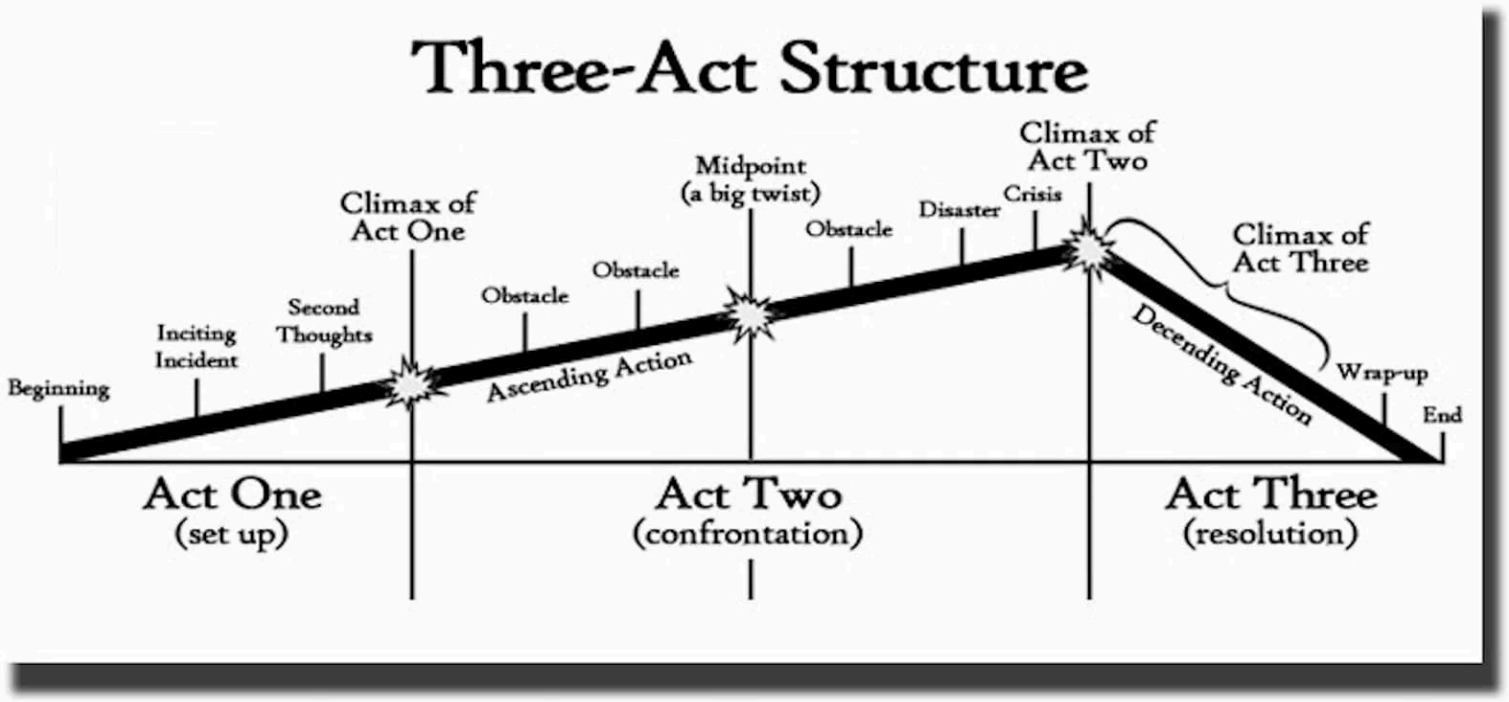
PART 02

Organizing a narrative sequence

Arrange information across views: build the explanation, control its pace, and preserve continuity.

Three acts establish, develop, and resolve

A **three-act narrative** divides a story into a setup, a development, and a resolution. Each act changes what the reader knows about the situation.



Act 1 · Setup

Introduce the setting and a question or tension.

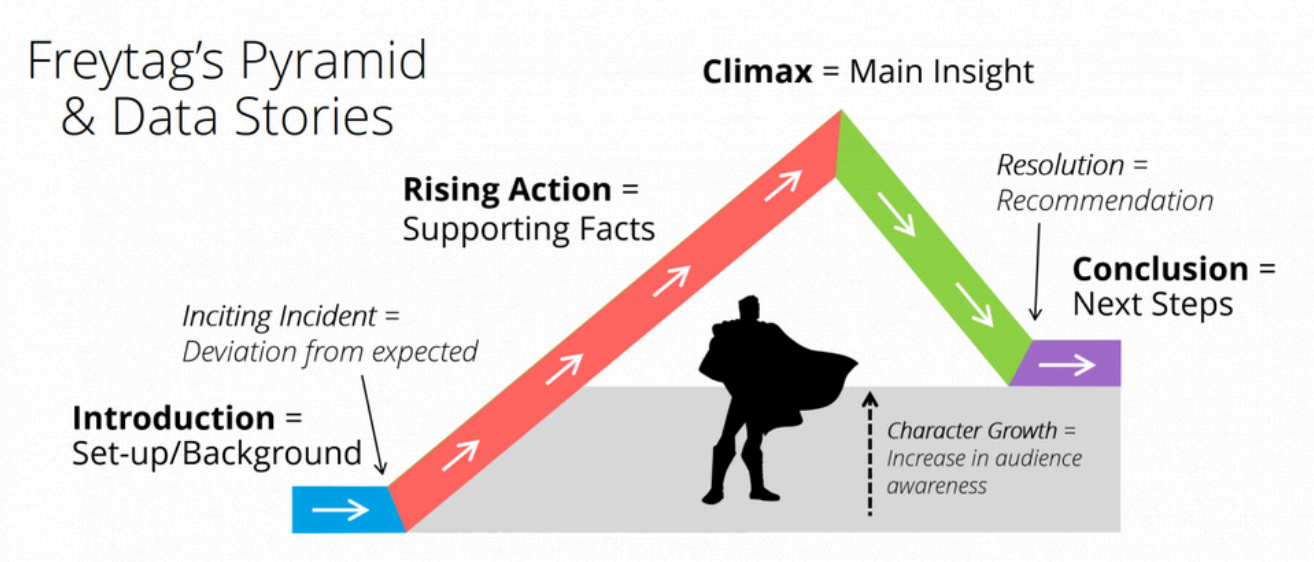
Act 2 · Development

Develop that tension through events and obstacles: the diagram's **confrontation**.

Act 3 · Resolution

Bring the story to an outcome and show what it means.

Data stories can unfold in four moves



Main Insight: the answer

The central answer that emerges from the evidence and resolves the story's question.

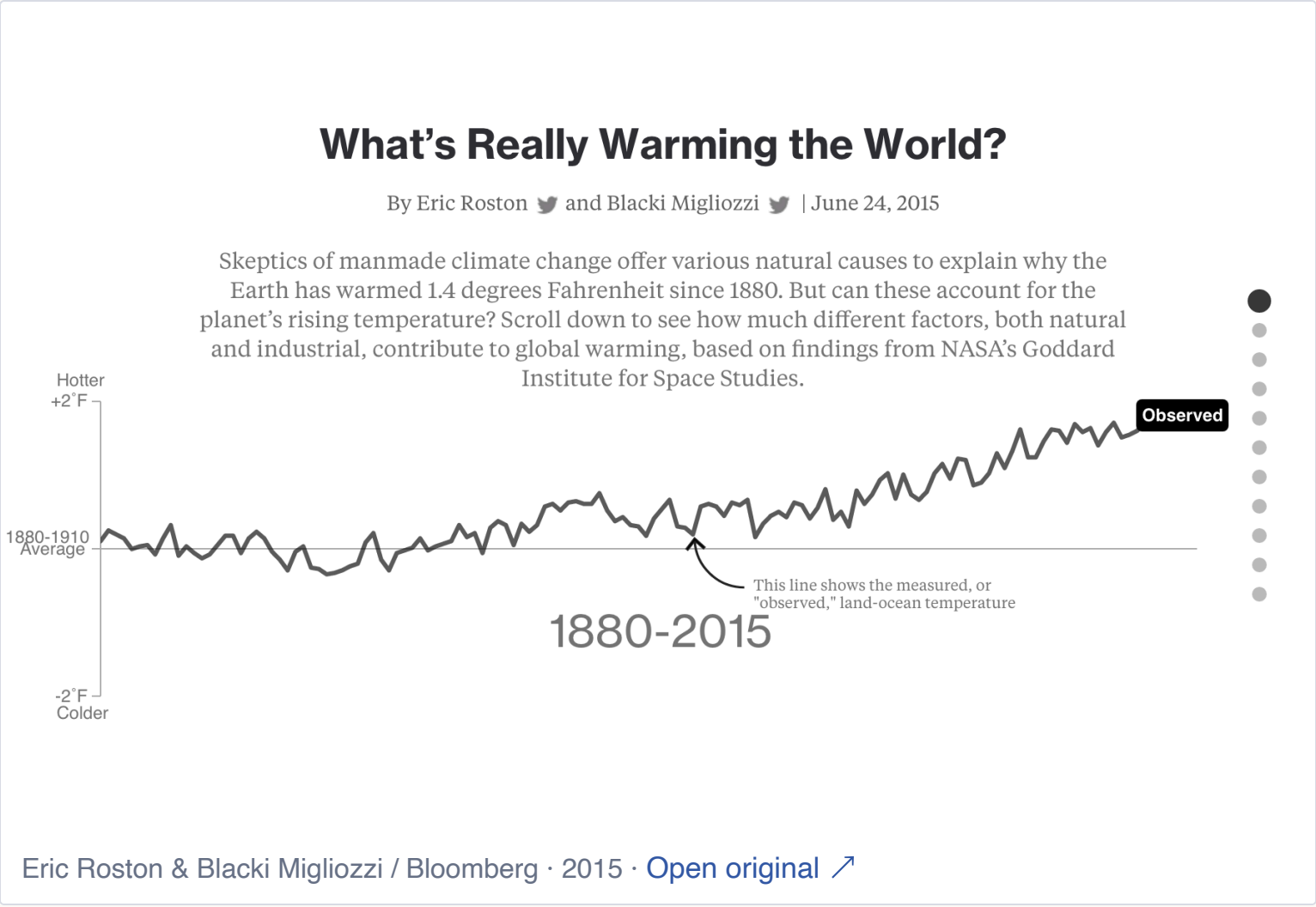
Implication: what it means

The answer's significance, consequences, or possible next steps.

1. Context	2. Question	3. Evidence	4. Implication
Introduction Set-up / Background	**Inciting Incident** Deviation from expected	**Rising Action → Climax** Supporting Facts → Main Insight	**Resolution + Conclusion** Recommendation / Next Steps

A teaching adaptation of the dramatic arc. An implication need not propose an action.

A sequence of questions about warming



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

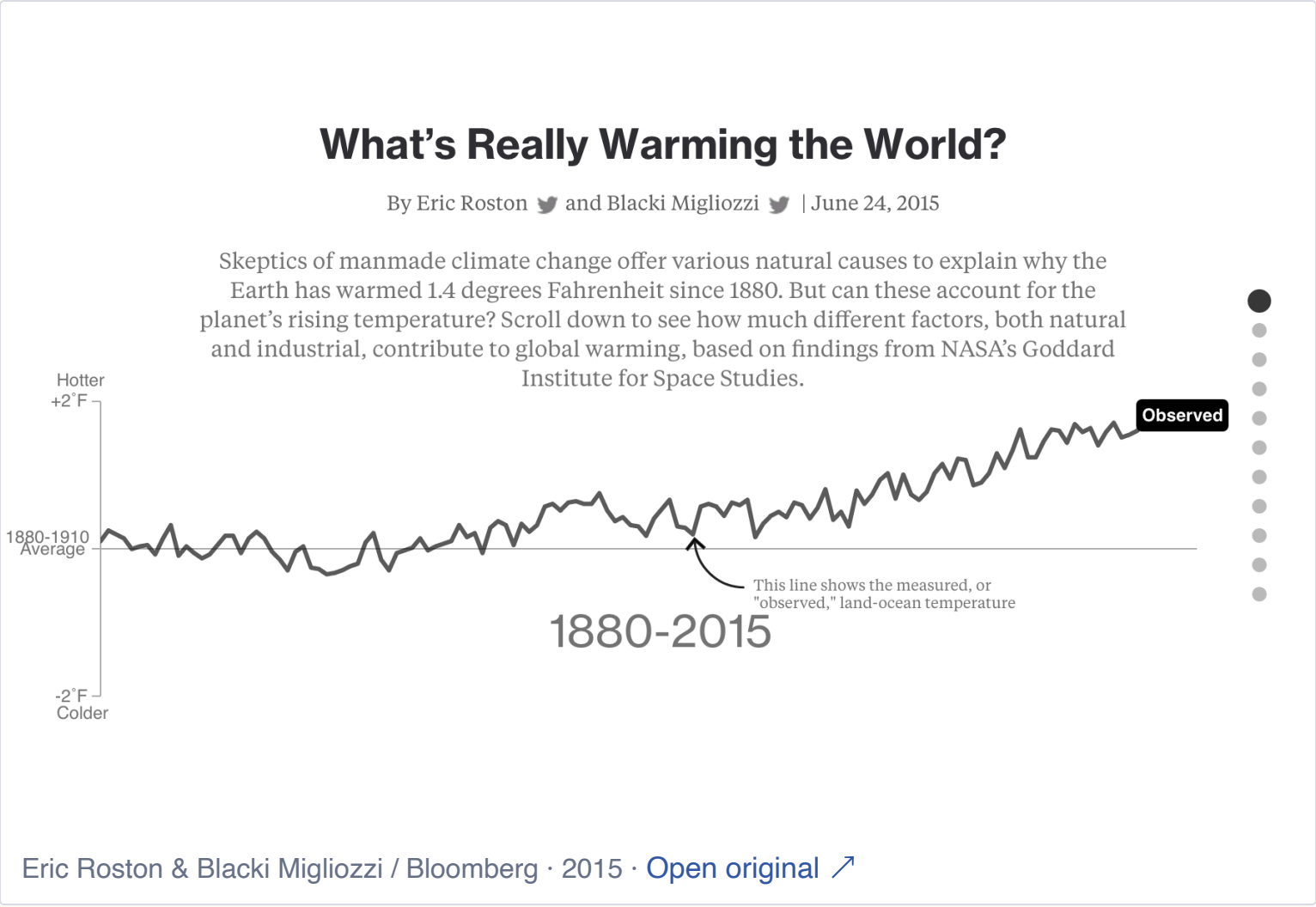
Follow the original

The opening introduces a temperature record and a question about its causes.

Later views consider different factors before bringing them together.

Watch the sequence: how do comparisons build toward an answer, and what does the ending ask next?

Context: start with the observed change



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

01 · CONTEXT

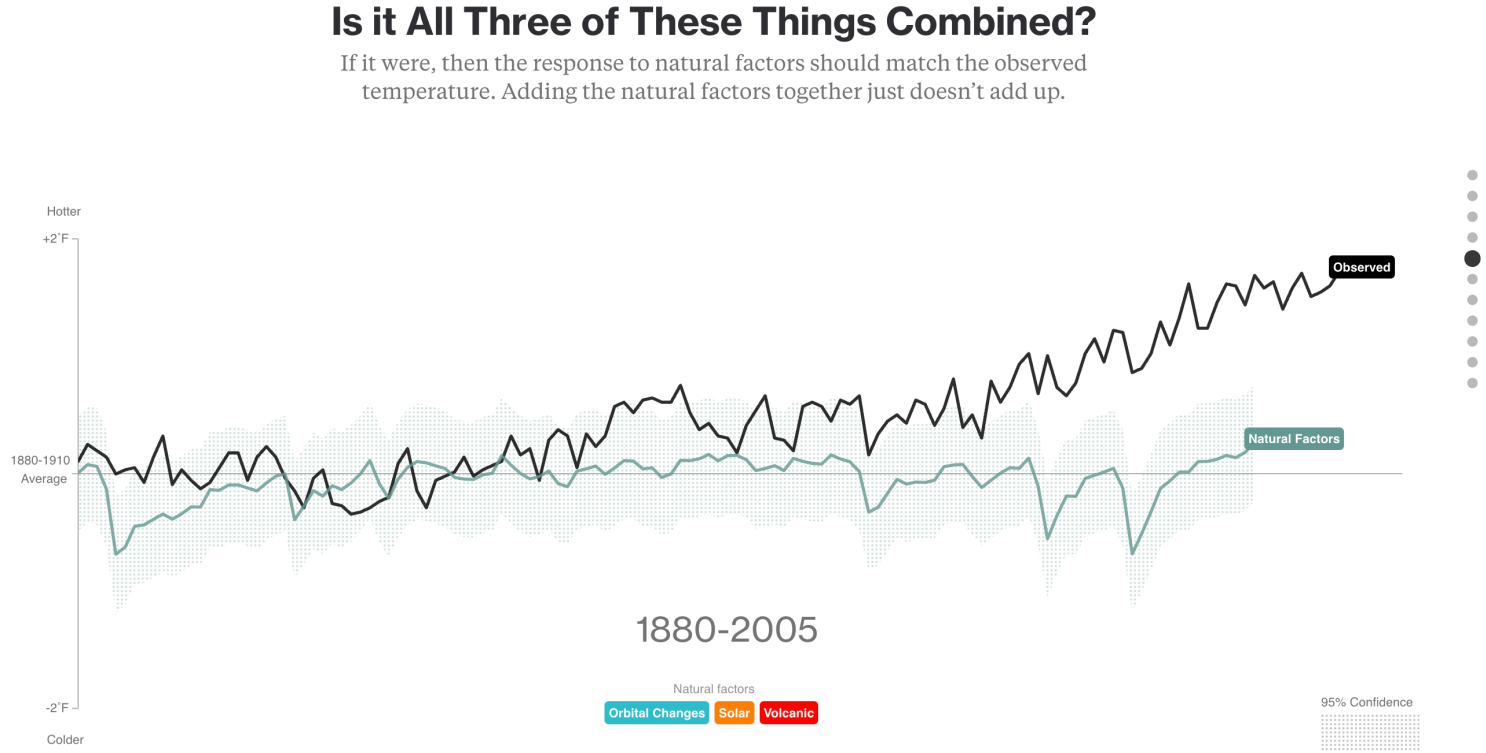
The observed record shows a long-term rise, especially toward the end of the period.

Readers meet the **observed temperature record** before any modeled factor appears.

This line becomes the reference for later views.

Question: could natural factors explain it?

Progressive disclosure introduces information in stages, making additional detail available as an explanation develops.



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

02 · QUESTION

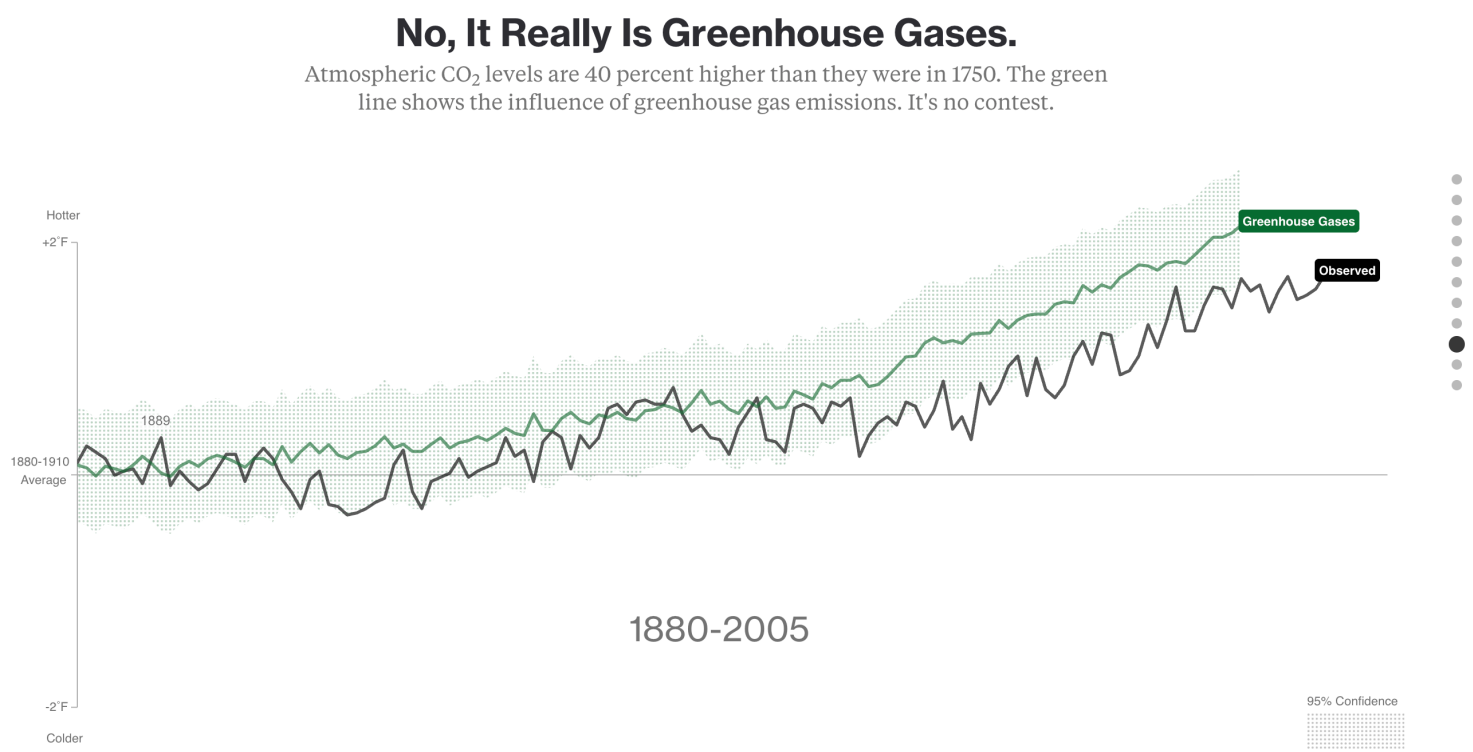
Natural factors enter first: orbit, solar variation, volcanoes, then their combined response.

Together, they do not reproduce the late-century rise.

Human factors enter in the following views, giving the explanation its next step.

Evidence: preserve continuity across comparisons

Object constancy is the perception that a visual object remains the same object as its appearance or surrounding view changes.



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

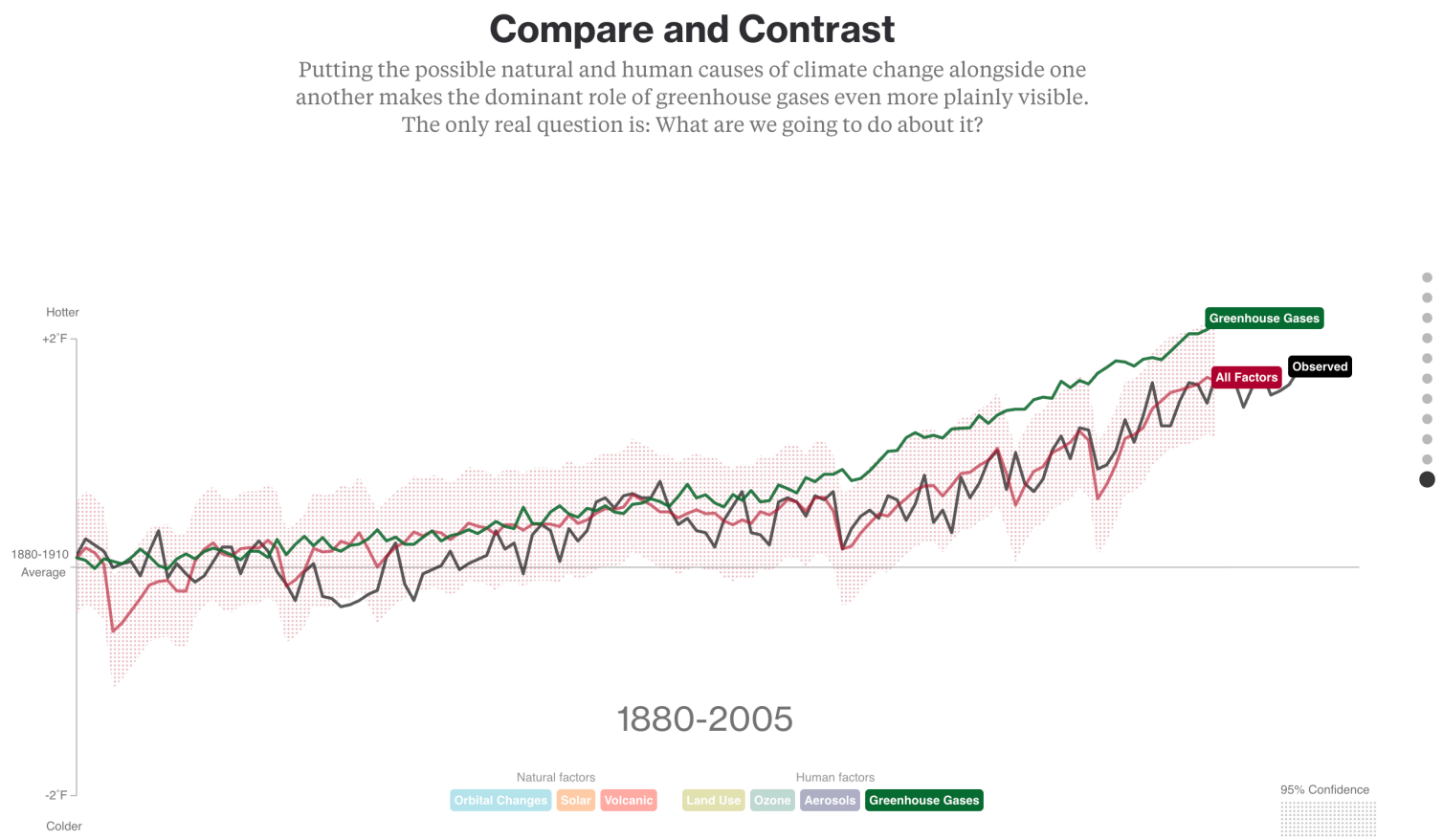
03 · EVIDENCE

The observed line and axes remain in place as the modeled factor changes.

Readers can carry the same reference into the greenhouse-gas comparison.

This new comparison builds toward the **main insight**.

Bloomberg: main insight and implication



Eric Roston & Blacki Migliozzi / Bloomberg · 2015 · Open original ↗

MAIN INSIGHT · THE ANSWER

Greenhouse gases are the dominant warming influence.

The combined model tracks the observed rise.

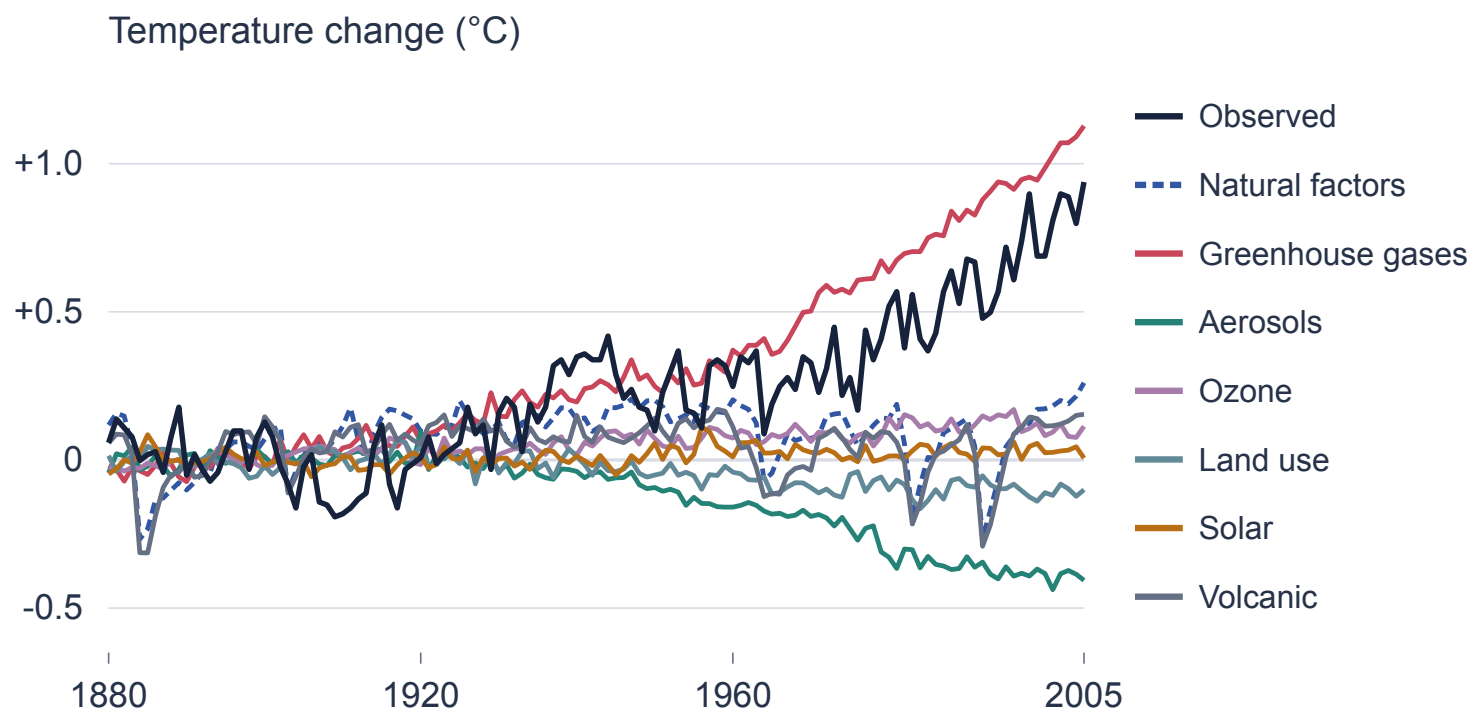
IMPLICATION · WHAT FOLLOWS

Attention turns to **how we respond**.

“What are we going to do about it?”

Bloomberg’s closing question

Pacing determines when each layer appears



Course replot · NASA GISS data distributed by Bloomberg · 1880–2005 · Open original ↗

Pacing is the timing and amount of information introduced at each stage of a story.

This view puts many factors on screen at once.

Progressive disclosure stages the comparisons; **object constancy** keeps the observed line as a familiar reference.

Compare: what would change if this were the first view?

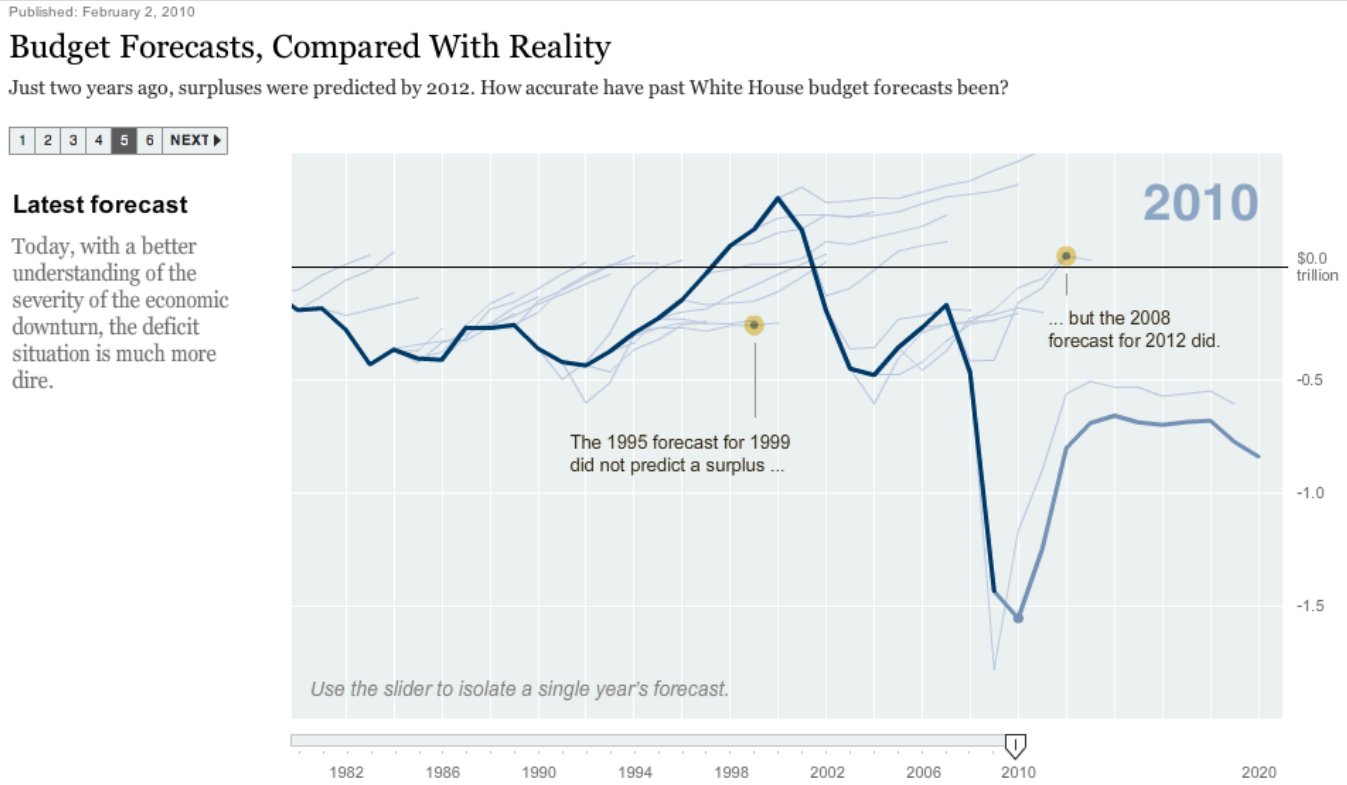
PART 03

Directing attention within a view

In a single static visualization, emphasis, annotations, and layout guide what readers notice and where they look next.

Visual emphasis makes a comparison stand out

Visual emphasis makes selected elements more noticeable through differences in color, contrast, size, or line weight, often by reducing the prominence of other elements.



The New York Times · 2010 · Figure 2 in Segel & Heer · Open archive ↗

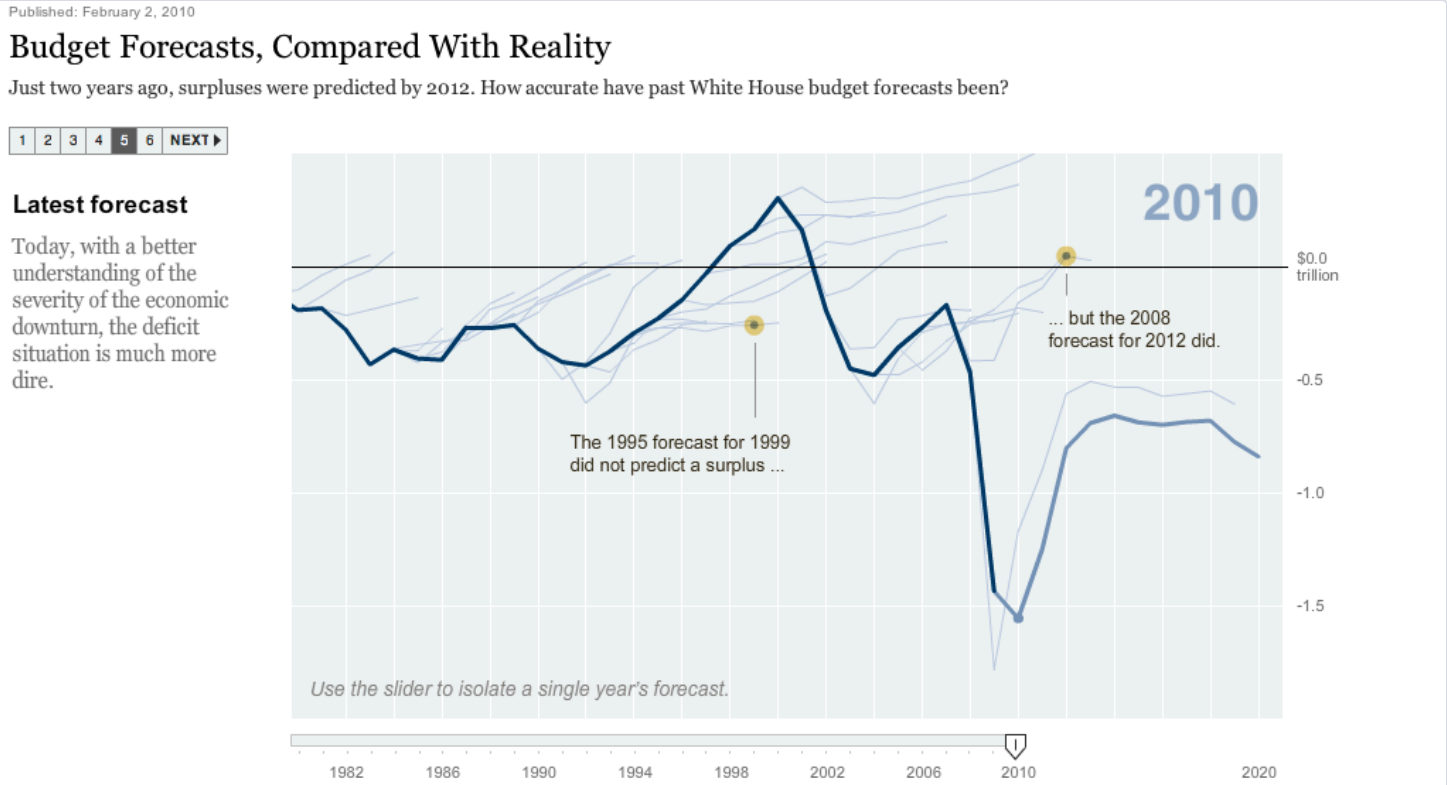
The dark, heavier **actual-results line** stands out from the pale forecasts.

Yellow markers draw attention to two comparison points.

The other forecasts remain visible as context.

Annotations explain the highlighted comparisons

An **annotation** is text or a visual cue attached to a specific part of a visualization to identify it, explain it, or add context.



The New York Times · 2010 · Figure 2 in Segel & Heer · Open archive ↗

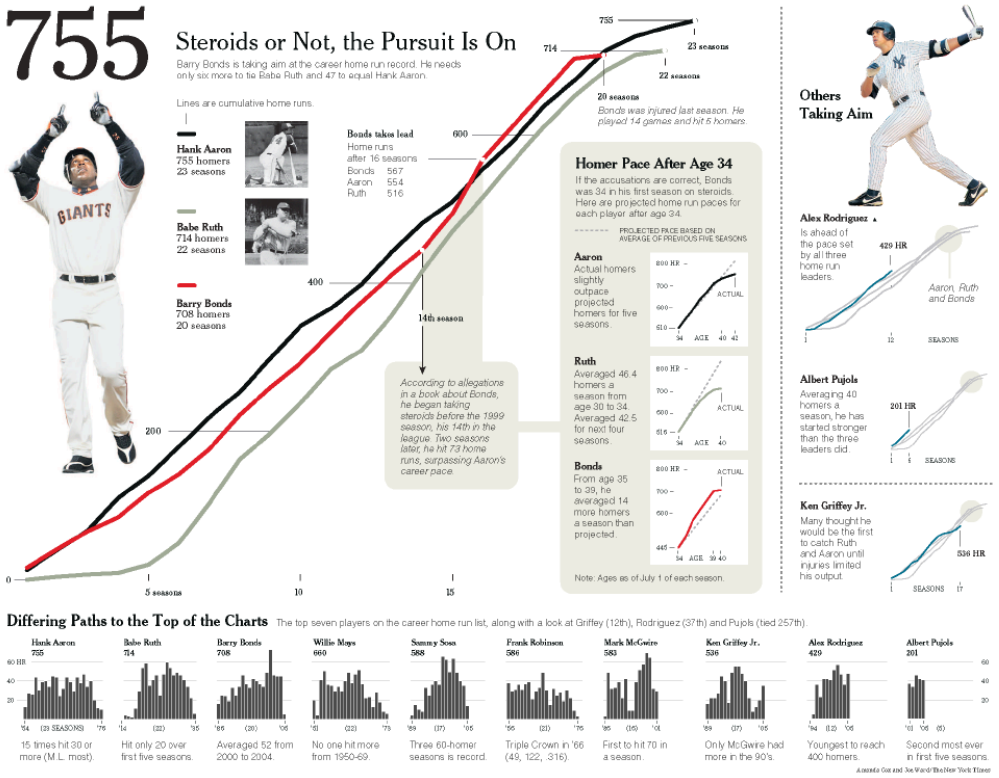
The **1995** and **2008** callouts describe how forecasts differed from later outcomes.

Lines and markers attach each explanation to its location in the chart.

Emphasis attracts attention; annotation gives the comparison meaning.

Layout suggests a reading path

A **reading path** is a suggested order for inspecting a graphic. Visual hierarchy, grouping, alignment, and connectors can guide readers through one static composition.



The New York Times · 2006 · Figure 1 in Segel & Heer · Read the case analysis ↗

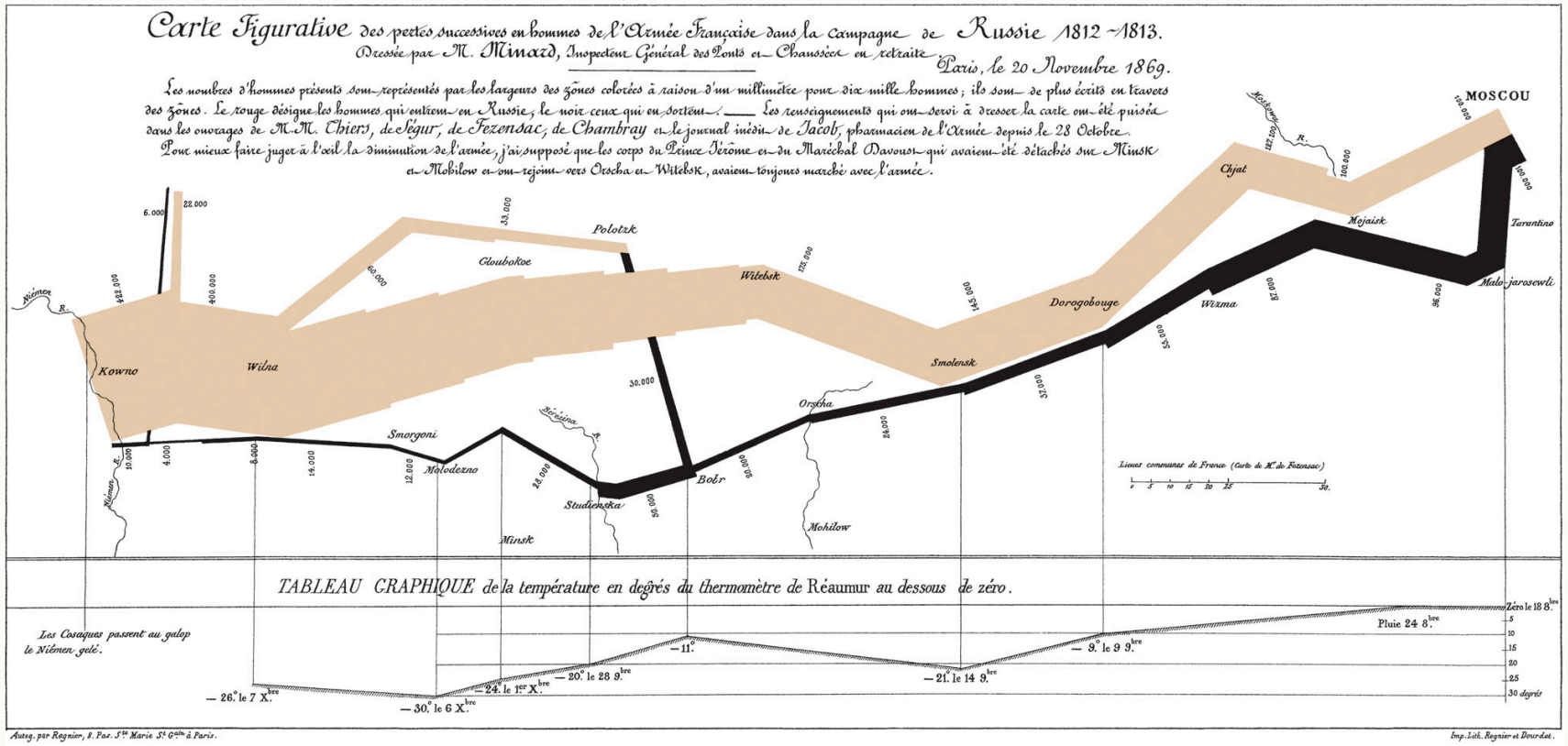
Enter: the large “755” and player figure.

Compare: the main career chart, then the inset linked by a shaded connector.

Extend: the smaller side and bottom panels.

The layout suggests a route; readers can choose another.

One static graphic can carry a narrative



Charles Joseph Minard · 1869 · Public domain · View original ↗

Emphasis
Narrowing bands reveal loss.

Annotation
Place names, counts, dates.

Reading path
Out → back → temperature.

PART 04

Balancing Explanation and Exploration

Use interaction to give readers room to explore within an authored explanation.

Guidance and exploration can support each other

Author-driven stories prescribe the sequence of scenes and provide substantial explanation. **Reader-driven** experiences let the reader choose paths, views, and details.

Visual cues guide **where readers look**. Narrative structure also determines **which comparisons readers can choose**.

Author guidance

Organizes an explanation and gives comparisons a shared context.

Earlier scenes prepare readers to interpret what comes next.

Reader exploration

Lets readers inspect details, choose comparisons, or follow a branch.

Readers pursue questions within the space the author provides.

The balance can change across a story. Guidance can prepare readers to explore.

Adapted from Segel & Heer (2010), Table 1 and §4.4.

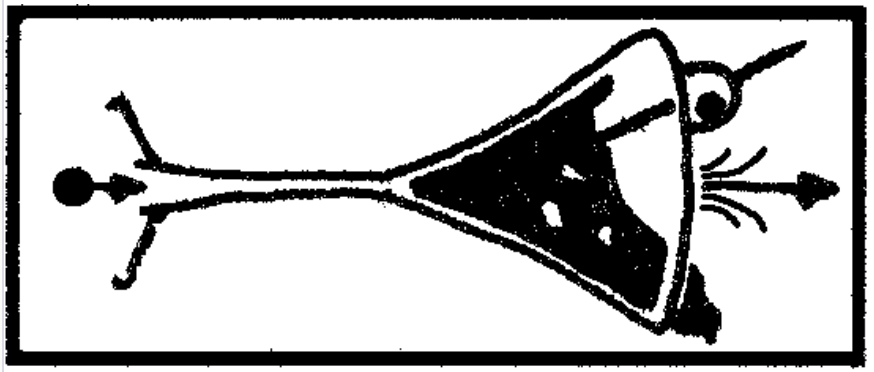
Martini glass: a guided opening, then exploration

A **martini-glass structure** starts with an authored sequence and opens into reader-directed exploration after that sequence.

Stem: a shared path introduces the topic and develops the explanation.

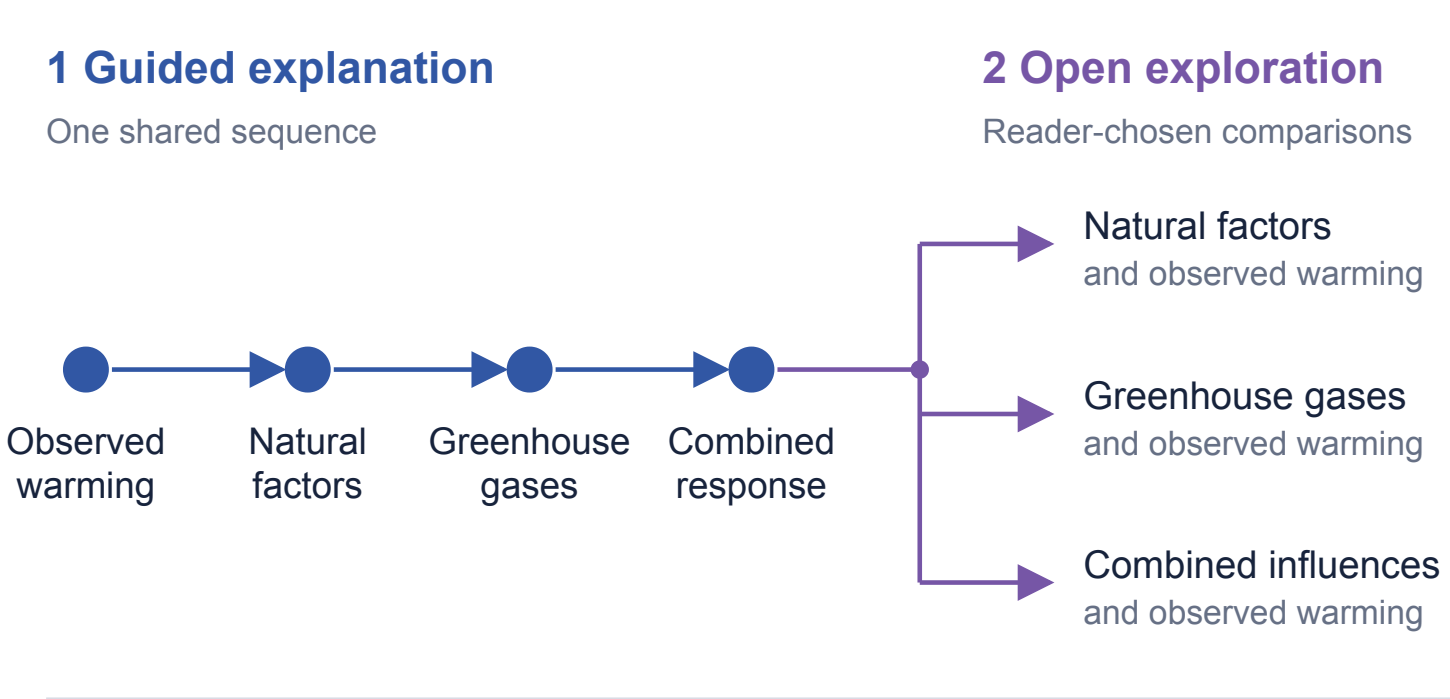
Bowl: readers choose among related views or comparisons.

The balance shifts from a shared explanation to reader-chosen comparisons. The opening prepares readers to explore.



Segel & Heer · 2010 · §4.4.1 · Read the paper ↗

Bloomberg guides first, then opens exploration



The guided sequence prepares readers for the comparisons they choose.

Course structural summary · Bloomberg, 2015 · Open original ↗

Guidance prepares exploration

First, the author guides: comparisons build toward a shared main insight.

Then, readers explore: they choose comparisons among those familiar factors.

The opening gives these later choices a shared context. The closing question asks how we respond.

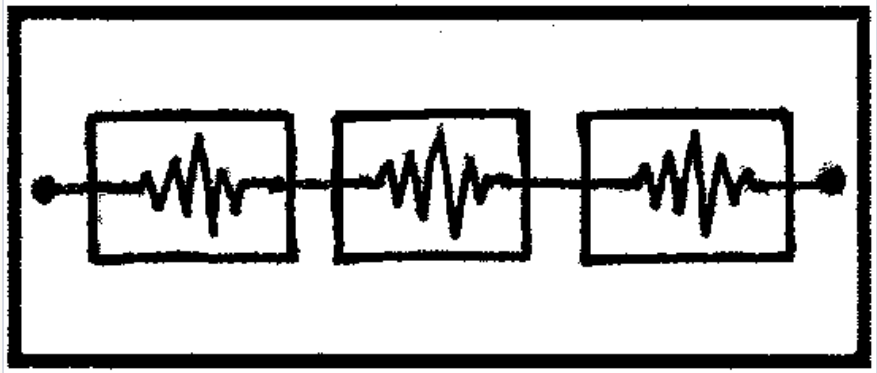
Interactive slideshows embed exploration in scenes

An **interactive slideshow** has an authored order of steps, with data interaction available inside individual steps.

Author guidance: the scene order develops the explanation.

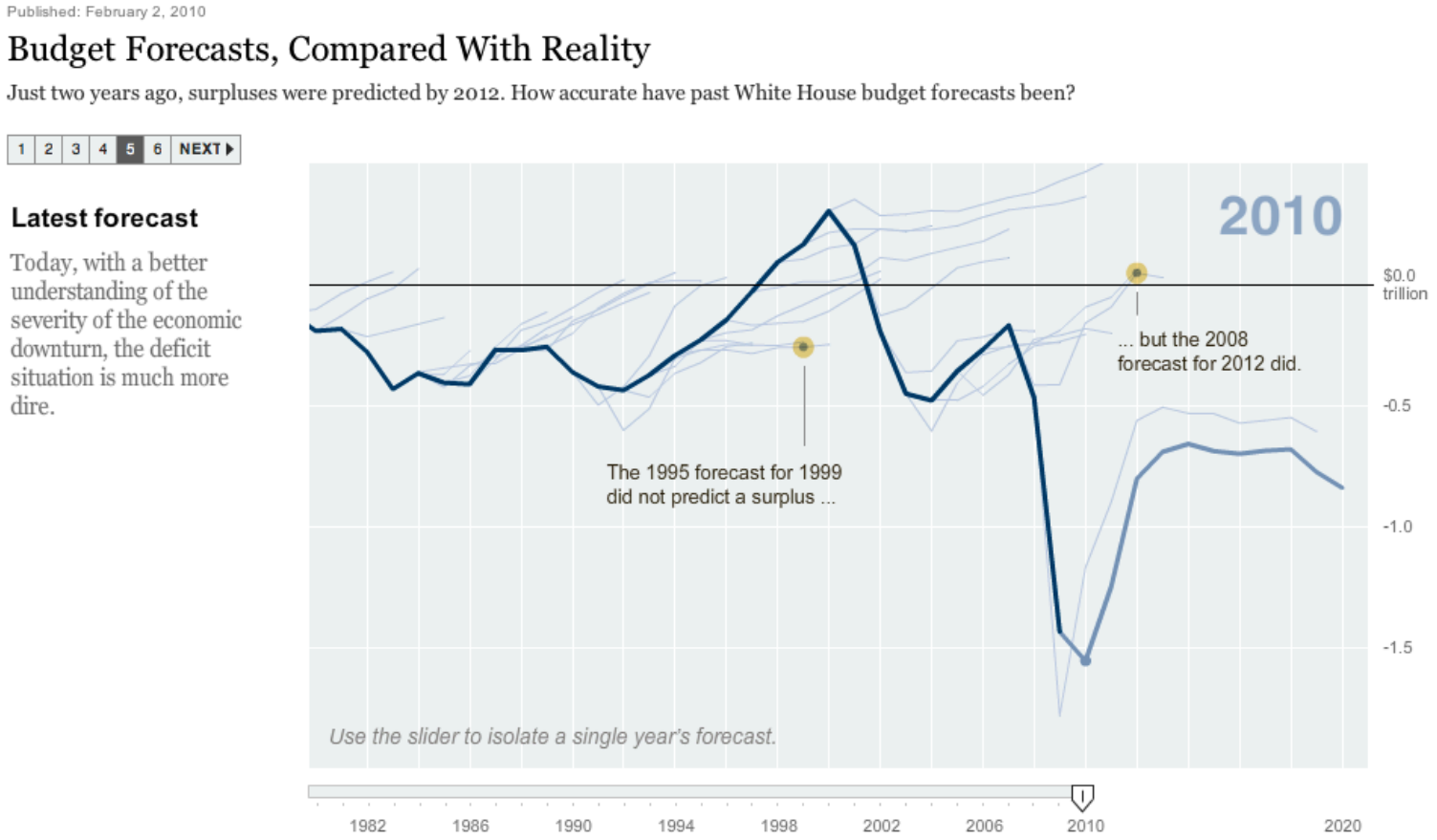
Reader exploration: choices inside a scene let readers investigate its comparison.

Readers return to the guided sequence with a deeper understanding of the current scene.



Segel & Heer · 2010 · §4.4.2 · Read the paper ↗

A local comparison deepens the current scene



The New York Times · 2010 · Figure 2 in Segel & Heer · Open archive ↗

A guided route with room to inspect

The author's numbered sequence explains how forecasts diverged from later outcomes.

Readers choose a forecast year to investigate that mismatch within the scene.

The next scene resumes the explanation, connecting the local comparison to the wider story.

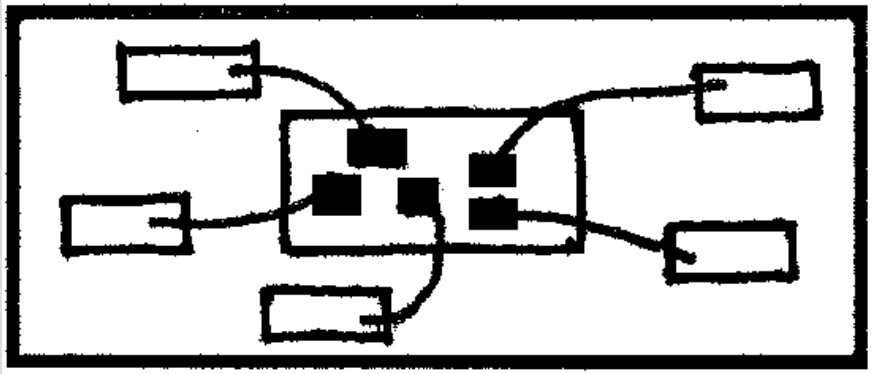
Drill-down stories let readers choose a case

A **drill-down story** introduces a general theme, then lets the reader choose particular cases to reveal more detailed accounts.

Author guidance: a shared overview frames the topic, and each branch provides an explanation.

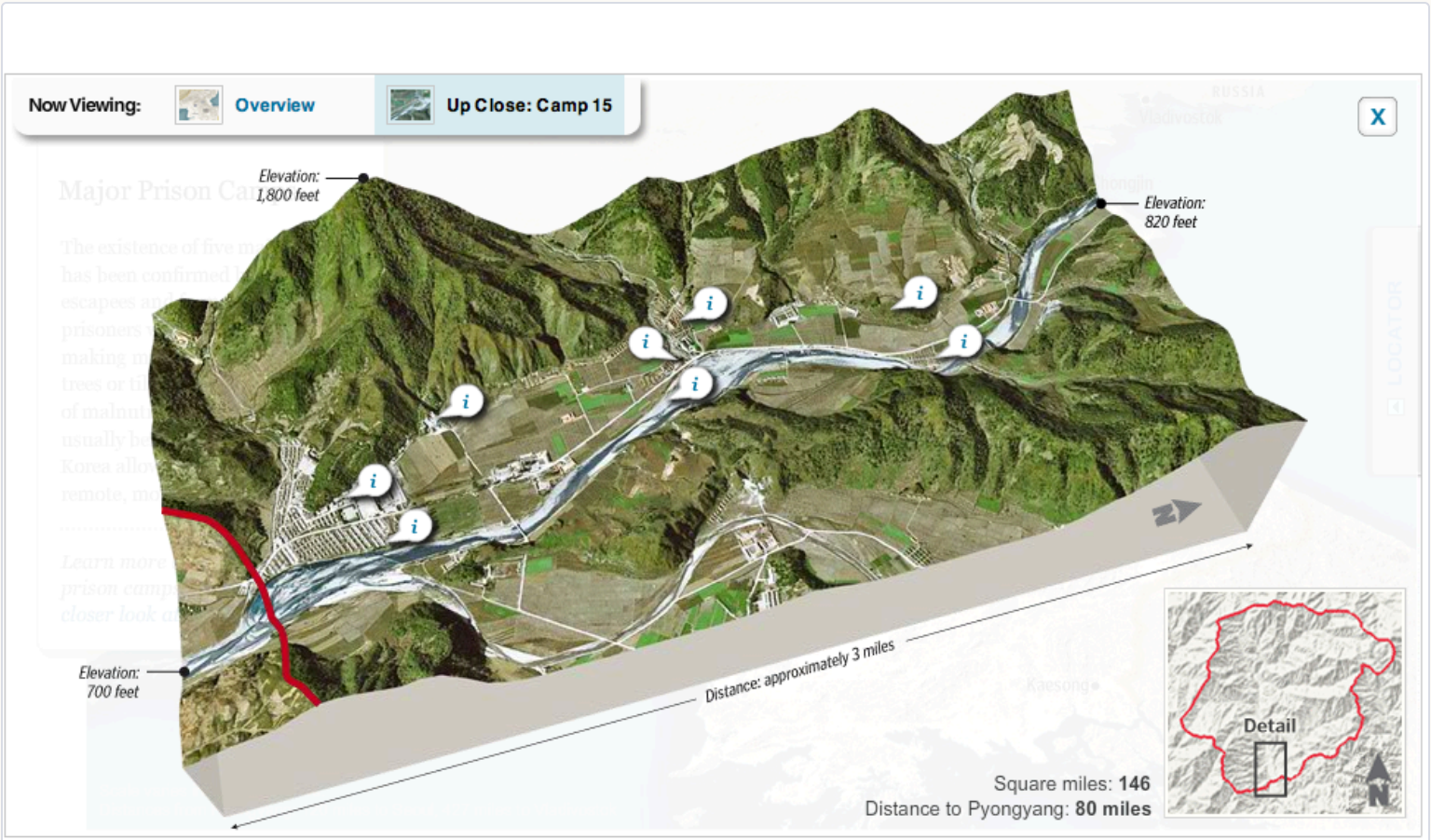
Reader exploration: readers choose which cases to visit and in what order.

The author provides the accounts, while readers assemble their route through them.



Segel & Heer · 2010 · §4.4.3 · Read the paper ↗

Readers choose paths through authored accounts



The Washington Post · 2009 · historical interface · Original project link ↗

Different paths through one theme

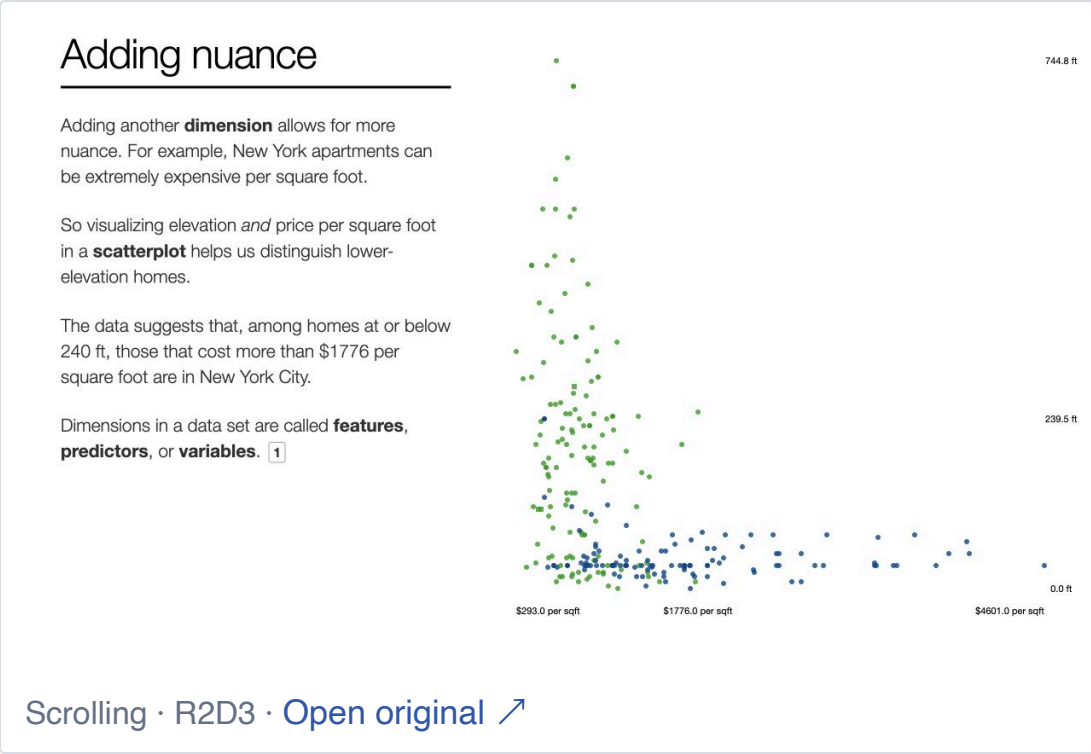
The author's overview establishes the wider topic of the camps.

Readers select a place and encounter an authored account of that setting.

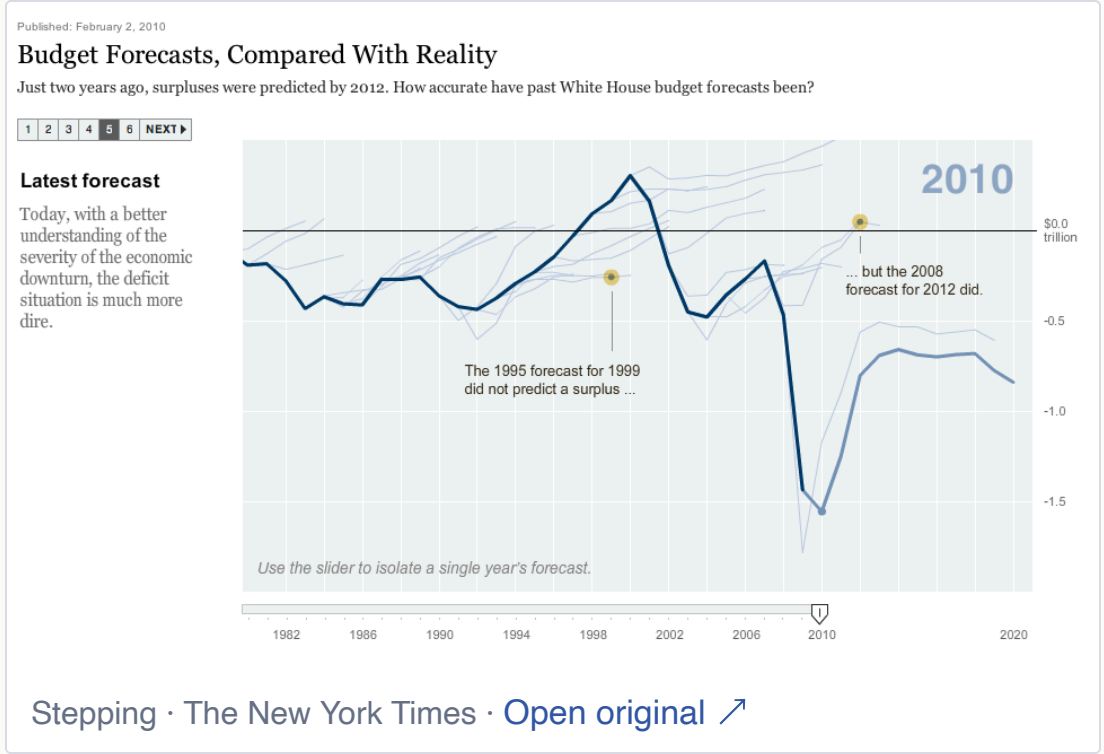
The shared theme connects the branches, while readers choose their route through them.

Pace can vary while the story order stays fixed

Readers control **when** they advance. Exploration also lets them choose **what** to inspect or compare.



Scrolling · R2D3 · Open original ↗



Stepping · The New York Times · Open original ↗

Scroll: the reader sets the pace as the visual explanation unfolds.

Discussion: Kosara (2016) · Bostock, How To Scroll

Stepper: the reader pauses within a scene before continuing.

Structure and genre describe different choices

Narrative structure describes how a story arranges author guidance and reader exploration across scenes and paths.

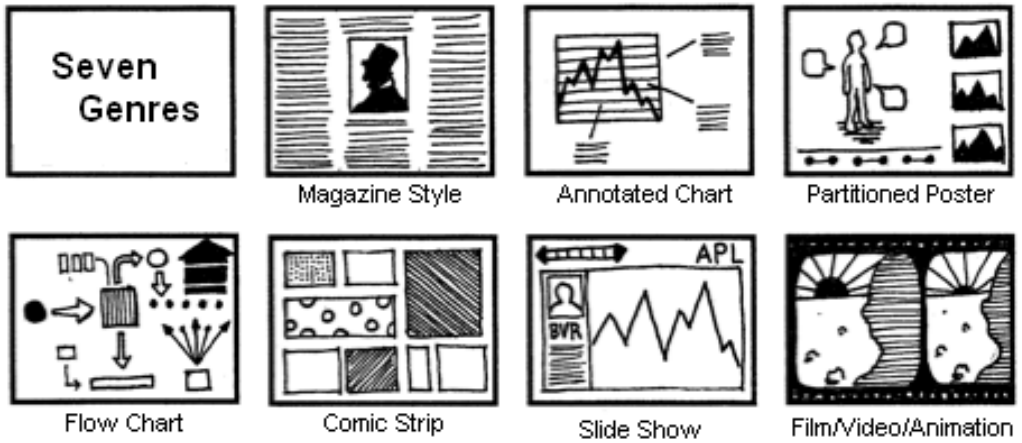
Visual genre describes the visual form used to present the story.

Choice	Examples
Structure	Martini glass · Interactive slideshow · Drill-down
Genre	Annotated chart · Magazine style · Comic strip · Film

A genre can support different balances of guidance and exploration. One structure can combine several genres.

Segel & Heer (2010), §§4.3–4.4.

Seven visual genres offer different forms



Segel & Heer · 2010 · Figure 8 · Read the paper ↗

Magazine style

Prose and graphics share a page.

Annotated chart

A chart carries attached explanations.

Partitioned poster

Related panels occupy one surface.

Flow chart

Connections guide a path or process.

Comic strip

Panels develop a visual sequence.

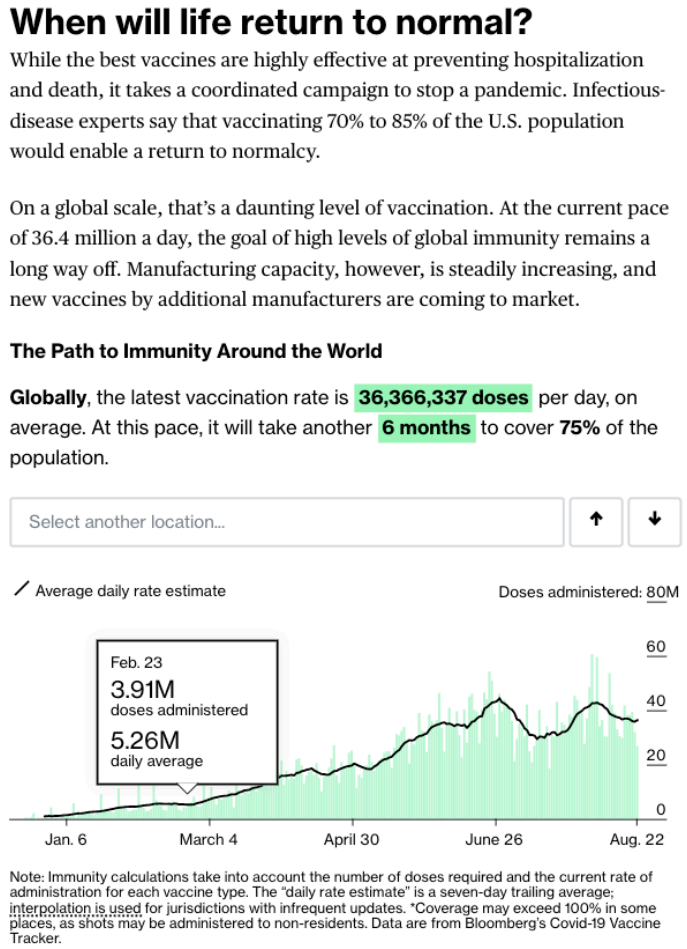
Slide show

Frames appear one after another.

Film / video / animation

Visual changes unfold over time.

Magazine style alternates prose and graphics



When will life return to normal?

While the best vaccines are highly effective at preventing hospitalization and death, it takes a coordinated campaign to stop a pandemic. Infectious-disease experts say that vaccinating 70% to 85% of the U.S. population would enable a return to normalcy.

On a global scale, that's a daunting level of vaccination. At the current pace of 36.4 million a day, the goal of high levels of global immunity remains a long way off. Manufacturing capacity, however, is steadily increasing, and new vaccines by additional manufacturers are coming to market.

The Path to Immunity Around the World

Globally, the latest vaccination rate is **36,366,337 doses** per day, on average. At this pace, it will take another **6 months** to cover **75%** of the population.

Select another location...

Average daily rate estimate

Doses administered: 80M

Note: Immunity calculations take into account the number of doses required and the current rate of administration for each vaccine type. The "daily rate estimate" is a seven-day trailing average; interpolation is used for jurisdictions with infrequent updates. *Coverage may exceed 100% in some places, as shots may be administered to non-residents. Data are from Bloomberg's Covid-19 Vaccine Tracker.

Bloomberg Vaccine Tracker · historical screenshot · Open original ↗

Magazine style integrates prose and graphics into the reading flow of an article.

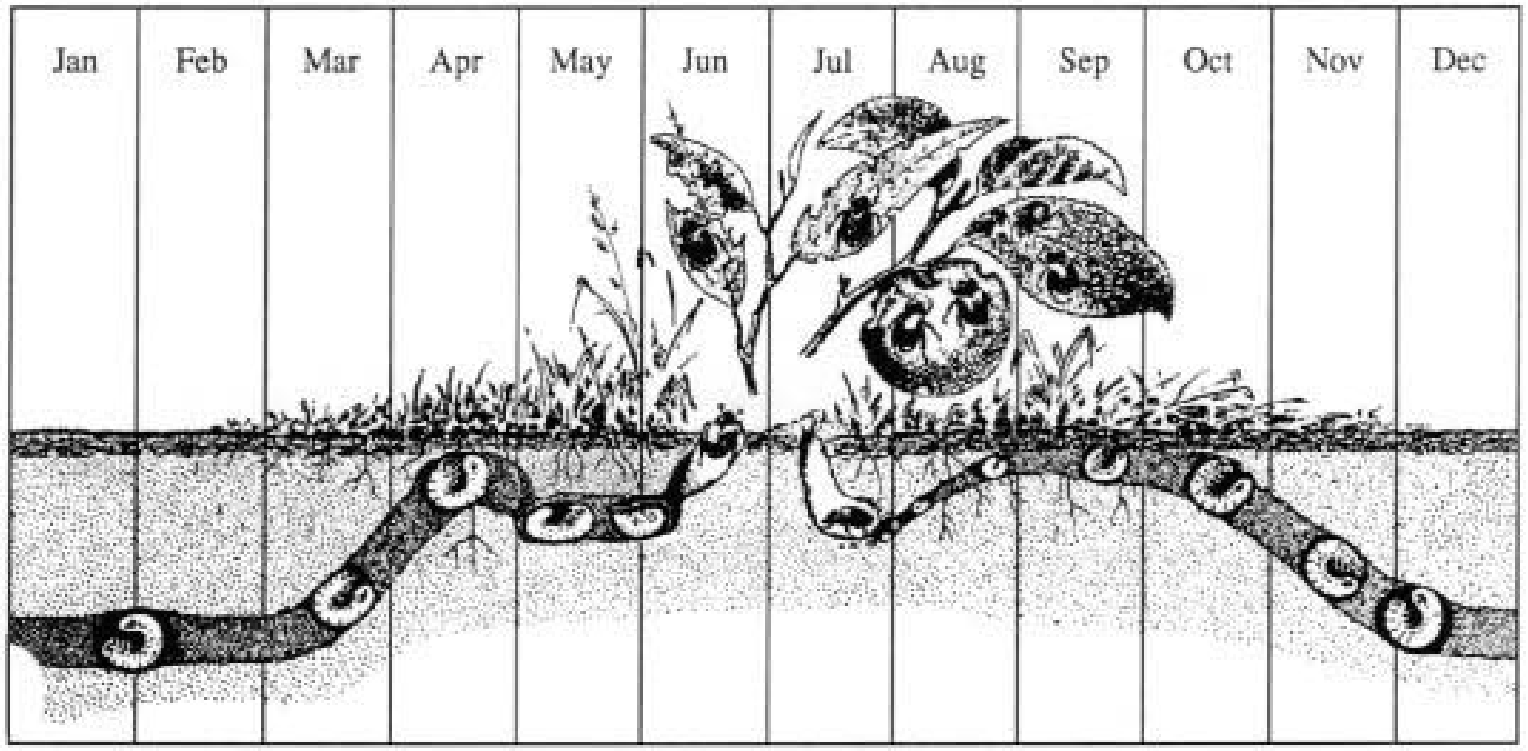
Paragraphs establish the topic and introduce the comparison.

The chart heading connects that discussion to the graphic below it.

The embedded chart and its note develop the explanation in the same column, keeping text and data together.

Historical publication state; the numbers and forecasts shown are not current.

Comic panels make time visible



Edward Tufte, The Visual Display of Quantitative Information, 2001; cited in Segel & Heer A43 · See the reference ↗

A **comic strip** develops a sequence through adjacent panels, combining images, labels, and changes between frames.

Here, month boundaries divide a continuous scene.

The author sets the order of stages. Readers choose how long to inspect each panel or compare it with another.

The communication task sets the balance

Communication task	Structure to consider	Possible visual form
Give a common introduction, then let readers compare	Martini glass	Article + annotated charts
Develop an explanation, with comparisons inside each scene	Interactive slideshow	Successive chart frames
Let readers enter through different places or cases	Drill-down story	Map + detail panels
Present a timed visual explanation	Author-driven sequence	Film / animation

Choose where readers can explore, then choose a visual form that supports that balance.

PART 05

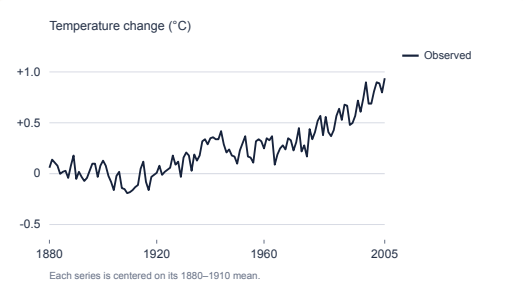
Storyboarding a data story

Bring sequence, visual emphasis, reader choices, and genre together in a plan for each scene.

A storyboard specifies each scene's design

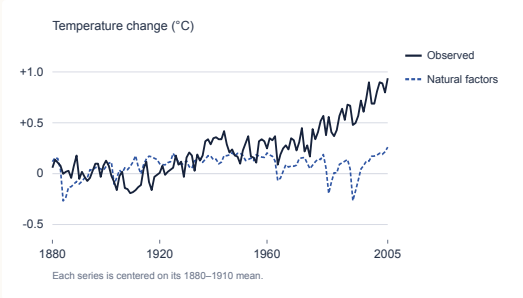
A **storyboard** is a sequence of rough frames that plans what each scene shows, what changes between scenes, and where readers can choose.

1 · CONTEXT



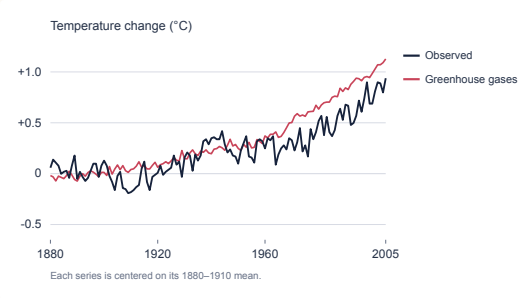
Show the observed line.
Annotate its late-century rise.

2 · QUESTION



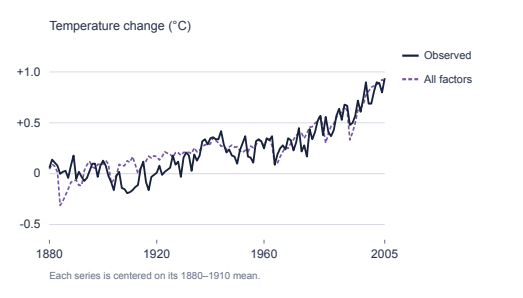
Keep axes and observed line.
Add natural factors and label the gap.

3 · EVIDENCE



Keep the reference.
Replace natural factors with greenhouse gases.

4 · IMPLICATION



Show the combined response.
State the answer, then ask how we respond.

Control: guide readers through frames 1–4, then open factor comparisons.

Storyboard plan using selected Bloomberg stages · reference thumbnails are course replots using NASA GISS data.

Cases and further reading

Cases

Hans Rosling / BBC: 200 years in 4 minutes

Bloomberg: What’s Really Warming the World?

R2D3: A Visual Introduction to Machine Learning

NYT: Budget Forecasts, Compared With Reality

Minard: the Russian campaign graphic (1869)

Washington Post: On the Map

Concepts and discussion

Segel & Heer (2010): Narrative Visualization

Seven genres, author/reader control, and three narrative structures.

Kosara (2016): The scrollytelling scourge

Examples and a critique of scroll-based progression.

Bostock: How To Scroll

Design and implementation considerations for scrolling.